VIT
Vellore Institute of Technology

Reg. No. : 22BAI1266

Final Assessment Test (FAT) - JUNE/JULY 2023

Programme	B.Tech.	Semester	Winter Semester 2022-23
Course Title	DIFFERENTIAL EQUATIONS AND TRANSFORMS	Course Code	BMAT102L
Faculty Name	Prof. ASHISH KUMAR	Slot	D1+TD1+TDD1
		Class Nbr	CH2022232300627
Time	3 Hours	Max. Marks	100

Part A (10 X 10 Marks)
Answer any 10 questions

01. a) Solve the differential equation $\frac{d^2y}{dx^2} + 16y = 8e^x$ by method of undetermined coefficients. **[5 marks]** [10]
b) Find the singular integral of $z = px + qy - 2\sqrt{(pq)}$. **[5 marks]**

02. An LCR circuit connected in series has $R = 10$ ohms, $C = 10^{-2}$ farad, $L = \frac{1}{2}$ henry and an applied voltage $E = 12$ volts. Assuming no initial current and no initial charge at $t = 0$ when the voltage is first applied. Find the subsequent current in the system at any time t without finding the charge. [10]

03. a) Solve by using Lagrange multiplier method $y^2p + x^2q = x^2y^2z^2$. **[7 marks]** [10]
b) Form a partial differential equation by eliminating the arbitrary constants a and b from $z = a(x + y) + b$. **[3 marks]**

04. Find the Laplace transform of $f(t) = tu(t-2) + \delta(t - \frac{\pi}{2})cost + \int_0^t \frac{sint}{t}dt$, where $u(t-2)$ and $\delta(t - \frac{\pi}{2})$ represent Unit step function and Impulse function respectively. [10]

05. Solve the given partial differential equation by using Laplace transform $\frac{\partial u}{\partial x} + x\frac{\partial u}{\partial t} = 0$ with boundary conditions $u(0,t) = t$ and $u(x,0) = 0$. [10]

06. Solve the given differential equation using Laplace transform [10]
$\frac{d^2y}{dt^2} - \frac{dy}{dt} = cos(2t) + cos(2t-12)u(t-6)$ with the boundary conditions $y(0) = 0$ and $y'(0) = 0$.

07. Find the half range Fourier cosine series of $f(x) = (\pi - x^2)$ in the interval $(0, \pi)$. Hence find the sum of the infinite series $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \ldots$ by using Parseval's identity. [10]

08. a) Determine the Fourier series of the periodic function $f(x) = x - 1$ defined on $(-\pi, \pi)$. **[5 marks]** [10]
b) Using convolution theorem find $f(t)$ if $L(f(t)) = \frac{1}{s(s^2+1)}$. **[5 marks]**

09. Evaluate $\int_0^\infty \frac{dx}{(x^2+4)(x^2+1)}$ using Fourier transform. [10]

10. Find the Fourier transform of $e^{-2|x|}$ and hence evaluate [10]

$$\int_0^\infty \frac{dx}{(x^2+4)^2}.$$

11. a) Find $Z[(n+1)^2e^{-5n+7}]$. **[5 marks]** [10]
b) Find $Z^{-1}[\frac{z^2}{(z-0.5)(z+3)}]$ using convolution theorem. **[5 marks]**

12. Solve $y_{n+2} - 3y_{n+1} + 2y_n = 3^n + 3$ with $y_0 = 1$ and $y_1 = 1$. [10]